

Extended E-R Model

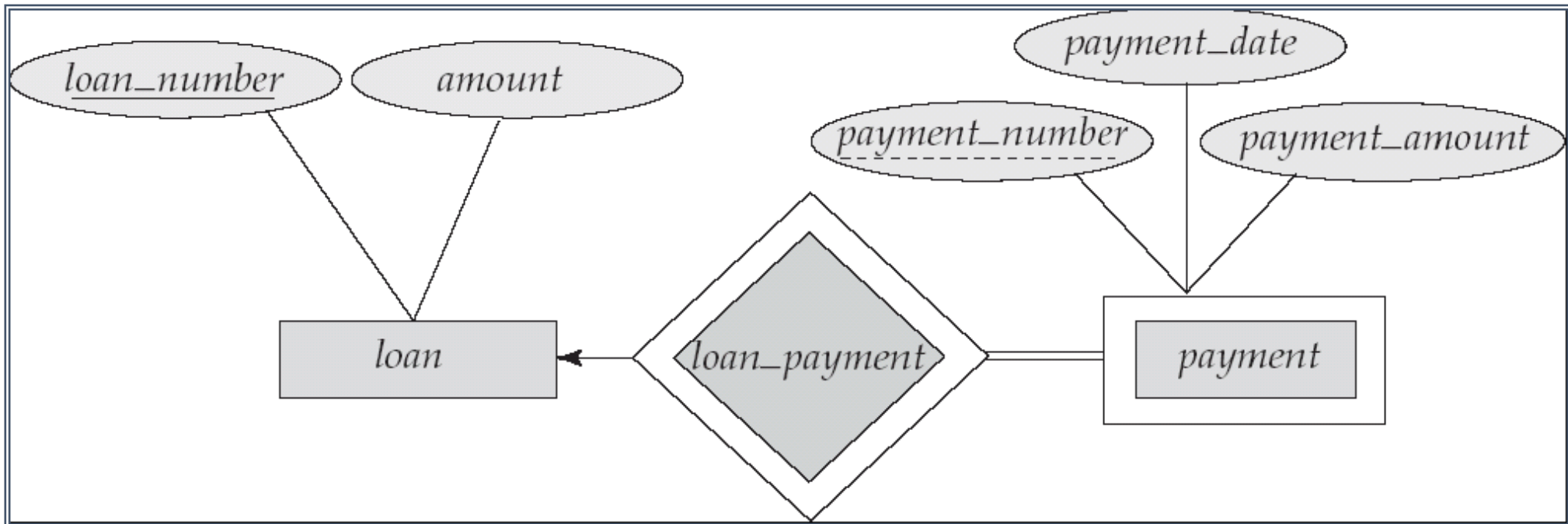
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Weak Entity Set

- ❑ An entity set that does not have a primary key is referred to as a **weak entity set**.
- ❑ The existence of a weak entity set depends on the existence of a identifying entity set
- ❑ It must relate to the identifying entity set via a total, one-to-many relationship set from the identifying to the weak entity set
- ❑ Identifying relationship depicted using a double diamond
- ❑ The **discriminator** (or **partial key**) of a weak entity set is the set of attributes that distinguishes among all the entities of a weak entity set.
- ❑ The **primary key of a weak entity set is formed by the primary key of the strong entity set** on which the weak entity set is existence dependent, plus the weak entity set's discriminator.

Weak Entity Set

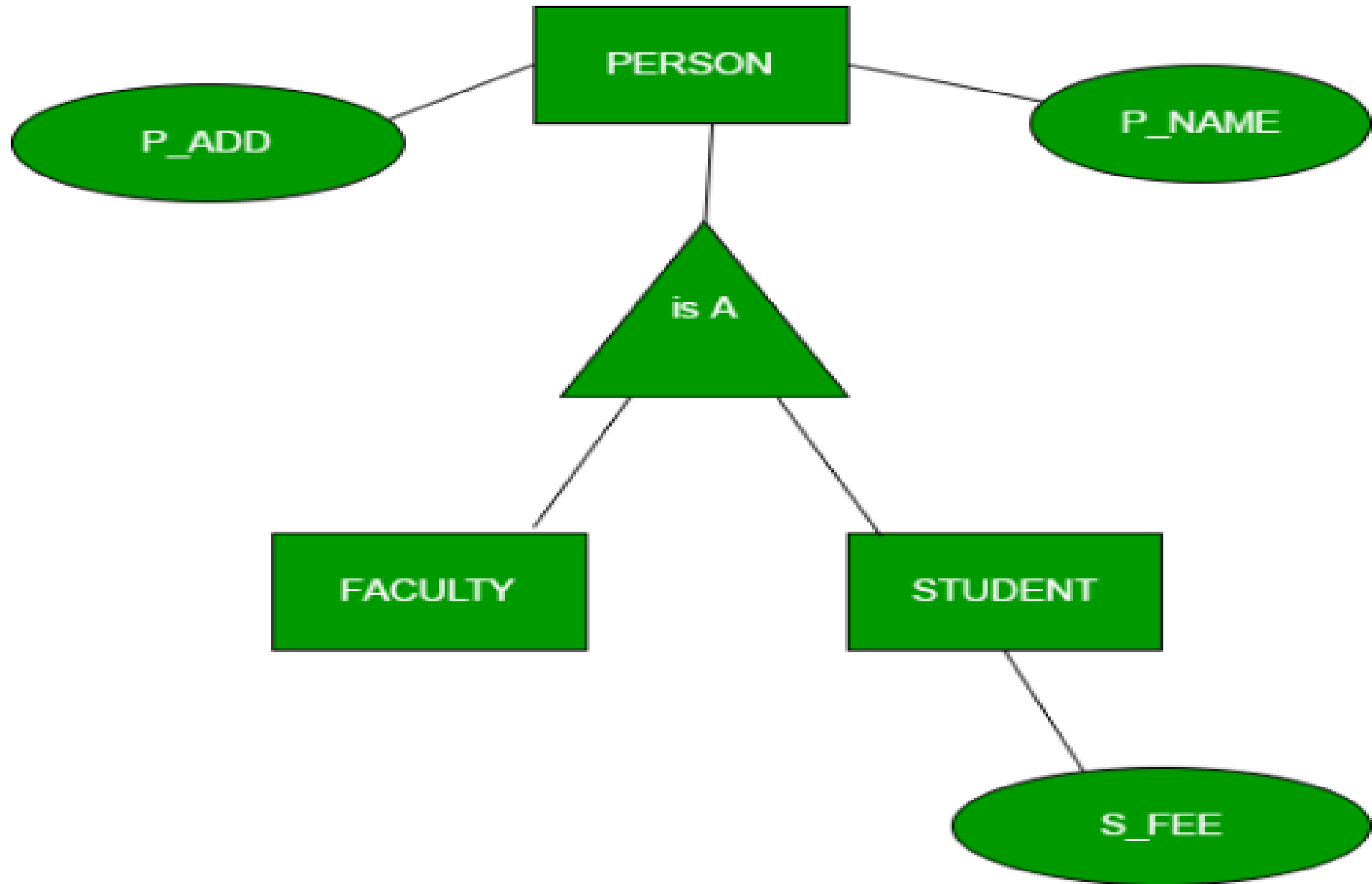
- ❑ We depict a weak entity set by double rectangles.
- ❑ We underline the **discriminator of a weak entity set** with a dashed line.
- ❑ `payment_number` – discriminator of the payment entity set
- ❑ Primary key for payment – (`loan_number`, `payment_number`)



Extended ER Diagram: Generalisation

Generalisation

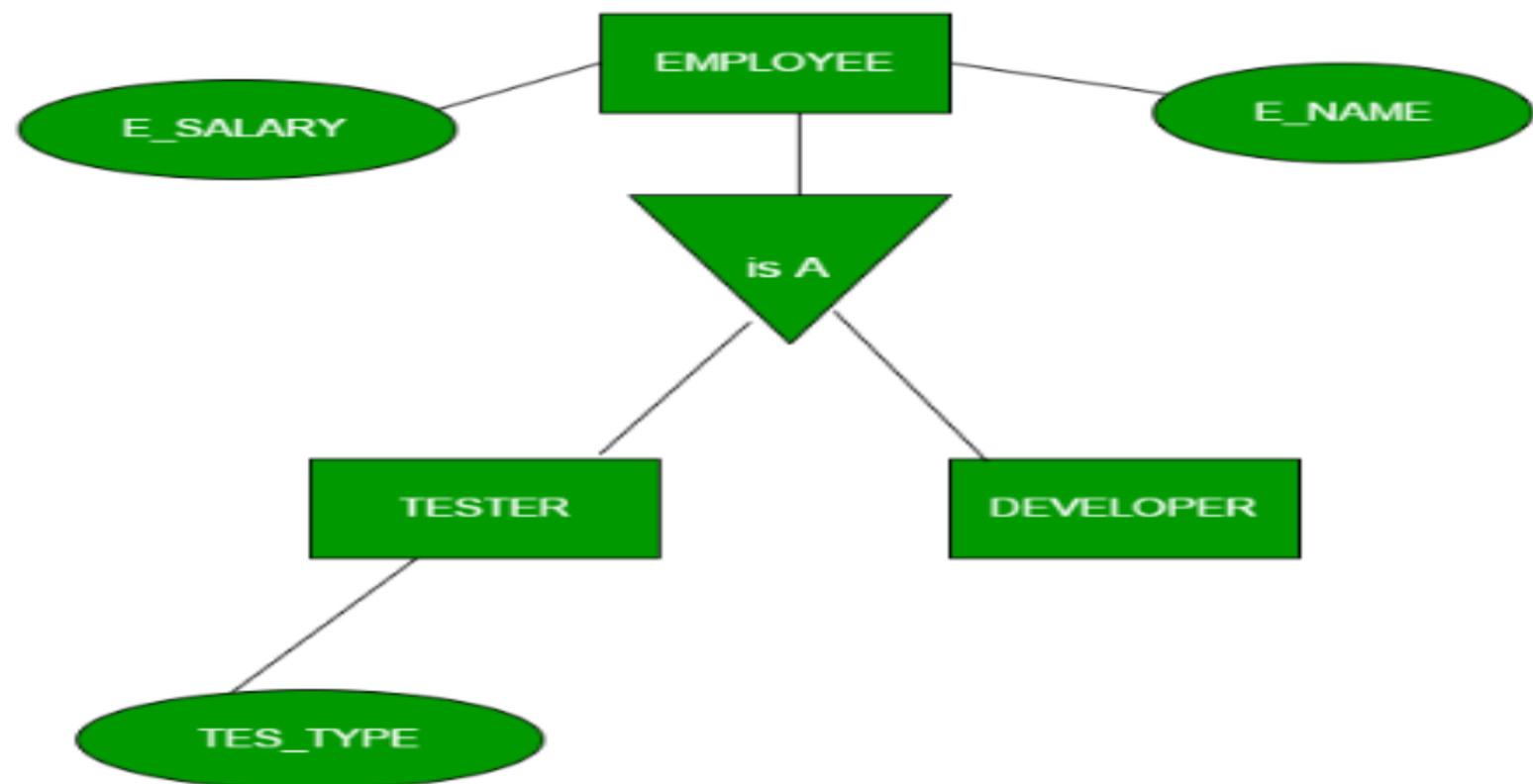
- ❑ Generalization is the process of extracting common properties from a set of entities and create a generalized entity from it.
- ❑ It is a bottom-up approach in which two or more entities can be generalized to a higher level entity if they have some attributes in common.
- ❑ For Example, STUDENT and FACULTY can be generalized to a higher level entity called PERSON as shown in Figure 1. In this case, common attributes like P_NAME, P_ADD become part of higher entity (PERSON) and specialized attributes like S_FEE become part of specialized entity (STUDENT)



Extended ER Diagram: Specialisation

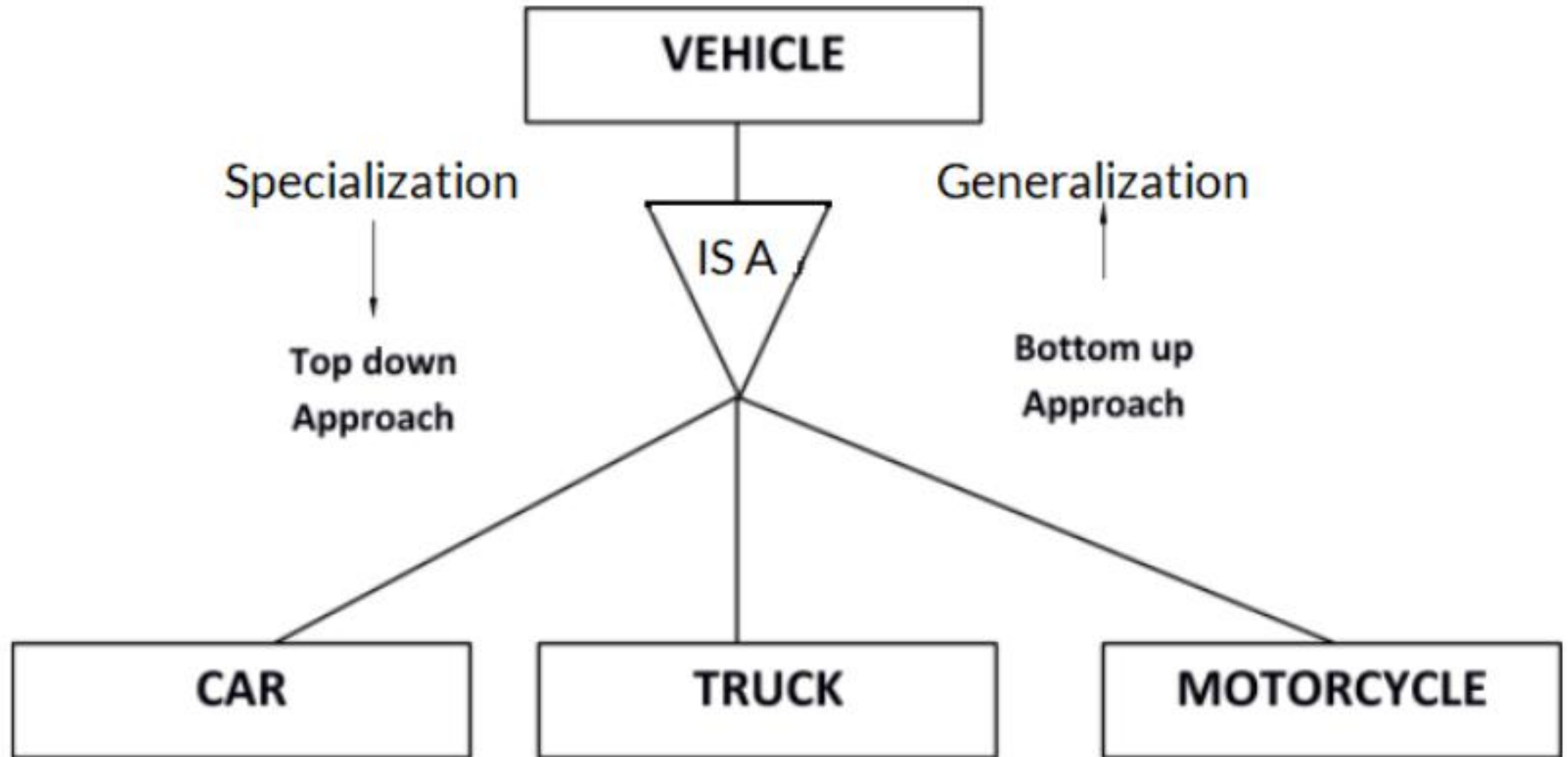
Specialization

- In specialization, an entity is divided into sub-entities based on their characteristics.
- It is a top-down approach where a higher-level entity is specialised into two or more lower-level entities.
- For Example, EMPLOYEE entity in an Employee management system can be specialized into DEVELOPER, TESTER etc. as shown in Figure 2.
- In this case, common attributes like E_NAME, E_SAL etc. become part of higher entity (EMPLOYEE) and specialized attributes like TES_TYPE become part of specialized entity (TESTER).
- Depicted by a triangle component labeled ISA (E.g. customer “is a” person).



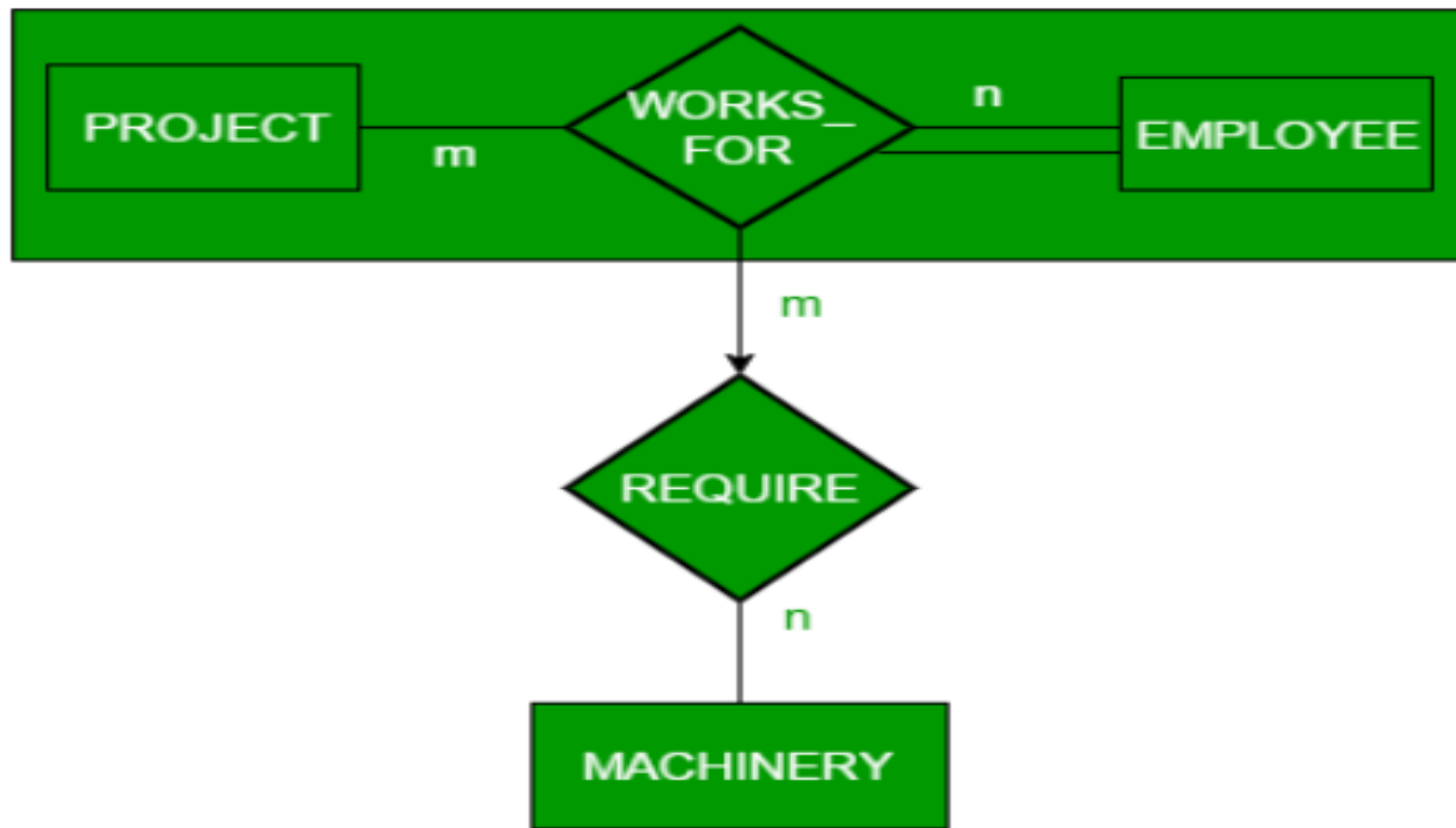
Specialization

Extended ER Diagram: Specialisation & Generalisation



Aggregation –

- An ER diagram is not capable of representing relationship between an entity and a relationship which may be required in some scenarios.
- In those cases, a relationship with its corresponding entities is aggregated into a higher level entity.
- Aggregation is an abstraction through which we can represent relationships as higher level entity sets.
- For Example, Employee working for a project may require some machinery. So, REQUIRE relationship is needed between relationship WORKS_FOR and entity MACHINERY. Using aggregation, WORKS_FOR relationship with its entities EMPLOYEE and PROJECT is aggregated into single entity and relationship REQUIRE is created between aggregated entity and MACHINERY.



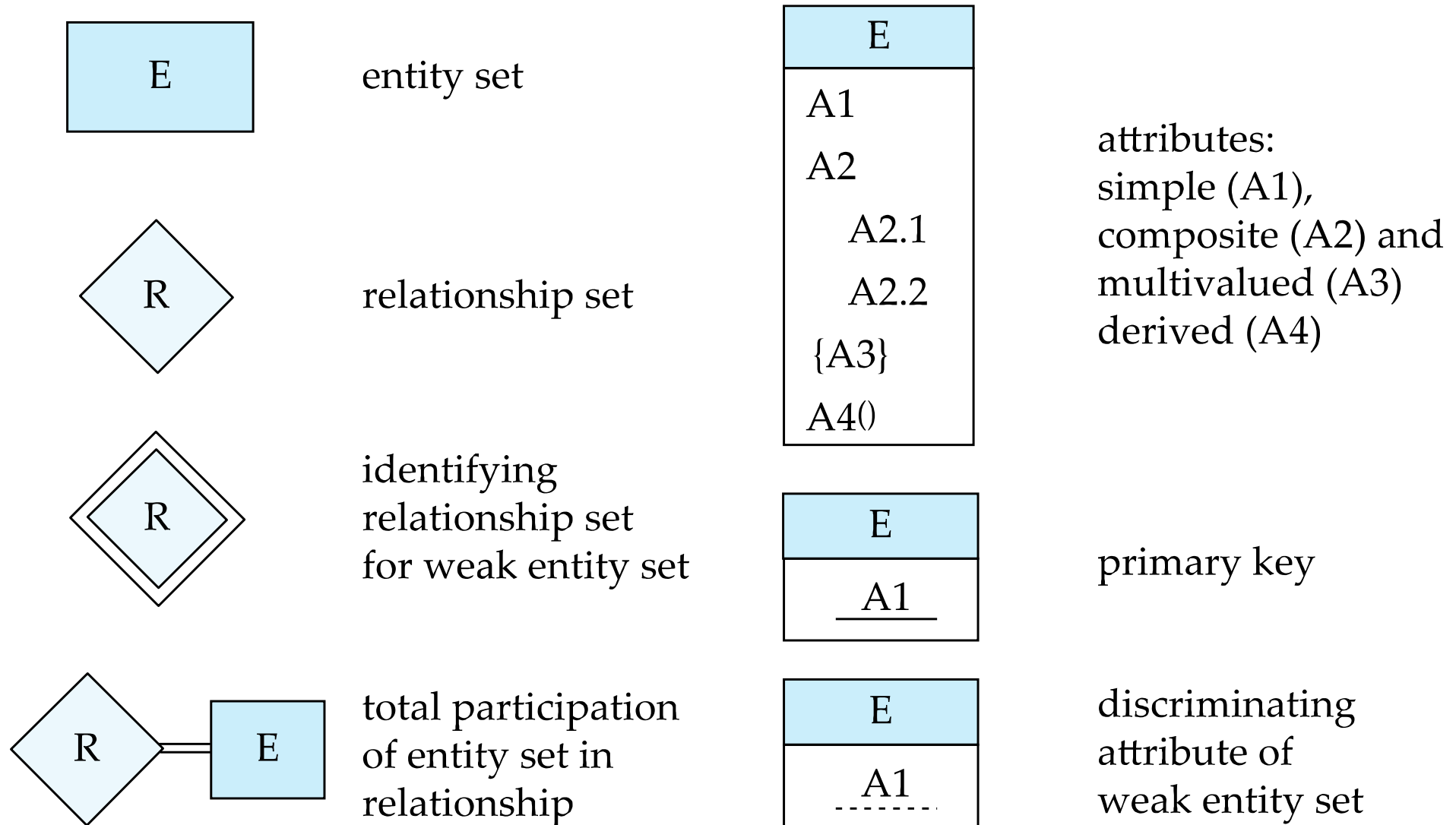
Aggregation

Representing aggregation via schema –

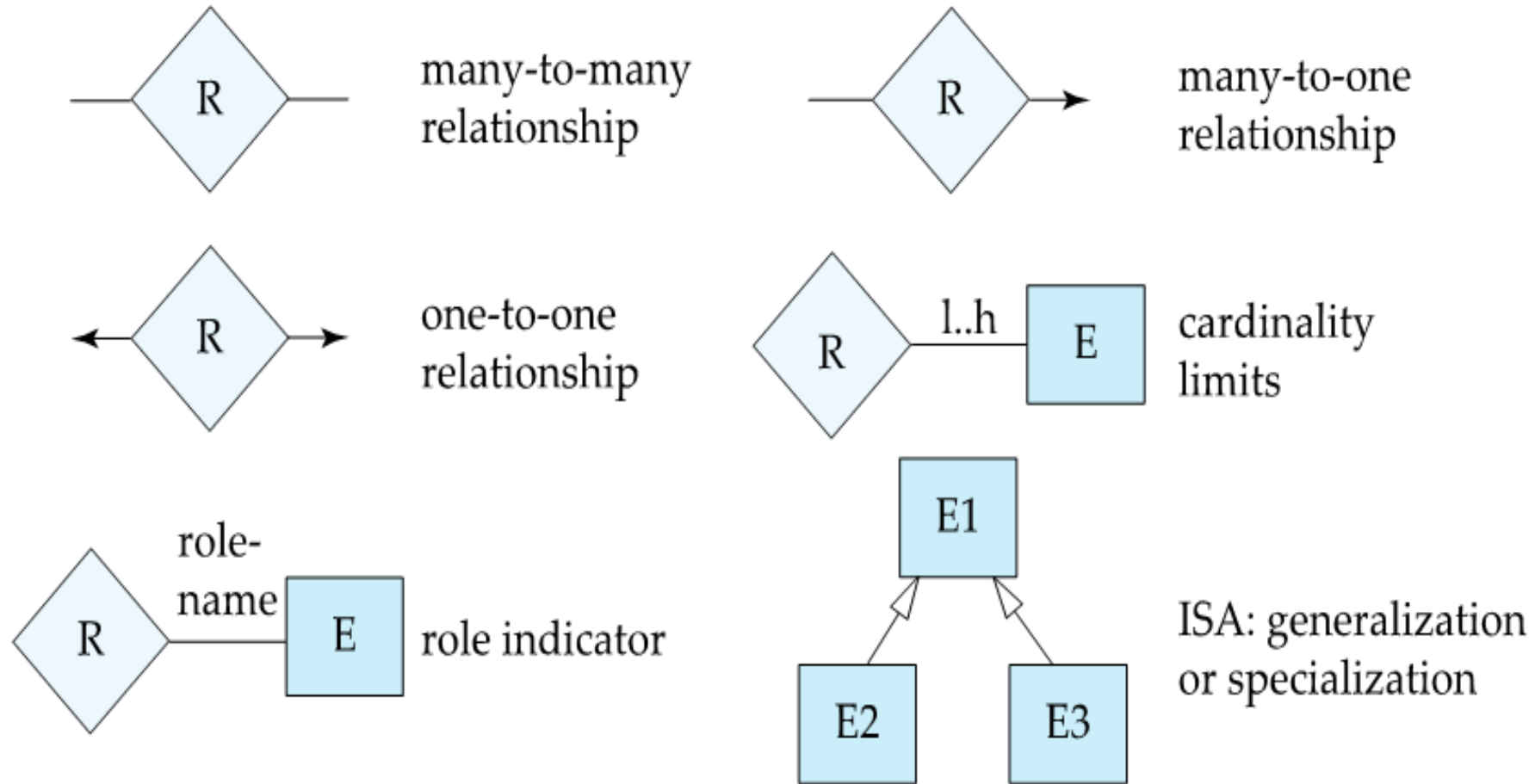
To represent aggregation, create a schema containing:

- 1.primary key of the aggregated relationship
- 2.primary key of the associated entity set
- 3.descriptive attribute, if exists.

Symbols used in E-R Diagram



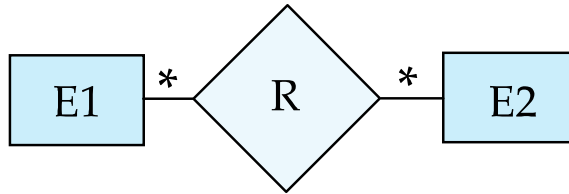
Symbols used in E-R Diagram (Cont.)



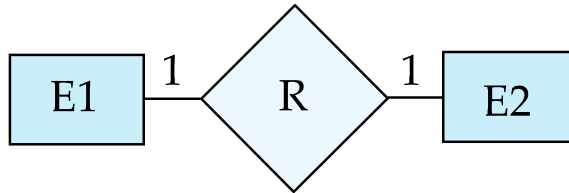
Symbols used in E-R Diagram (Cont.)

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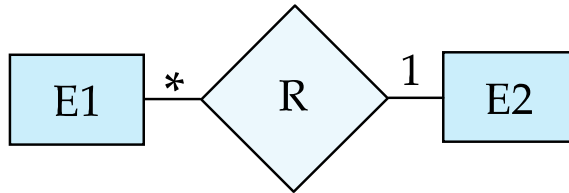
many-to-many
relationship



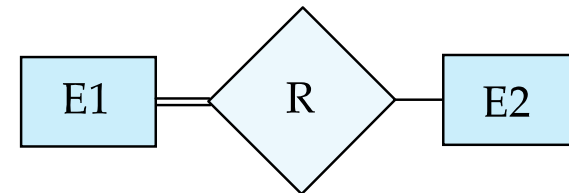
one-to-one
relationship



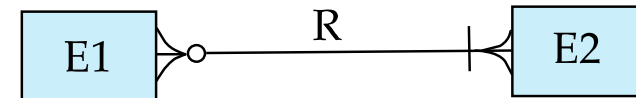
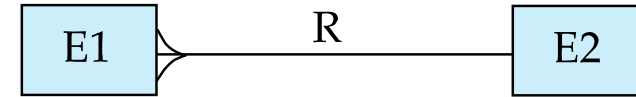
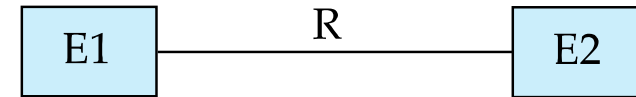
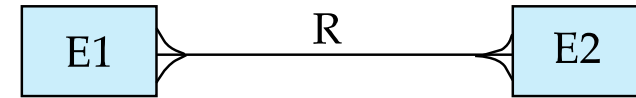
many-to-one
relationship



participation
in R: total (E1)
and partial (E2)



IDE1FX (Crows feet notation)



Reducing ER Diagrams to Tables

- ❑ Rule 1 : Strong Entity Set with Simple attributes
- ❑ Rule 2 : Strong Entity Set with Composite attributes
- ❑ Rule 3 : Strong Entity Set with Multi Valued Attributes
- ❑ Rule 4 : Binary relationship with cardinality ratio M:N
- ❑ Rule 5 : Binary relationship with cardinality ratio 1:N
- ❑ Rule 6 : Binary relationship with cardinality ratio N:1
- ❑ Rule 7 : Binary relationship with cardinality ratio 1:1
- ❑ Rule 8 : Binary relationship with Weak Entity Set
- ❑ Rule 9 : Binary relationship with total participation constraint
- ❑ Rule 10: N-ary relationship

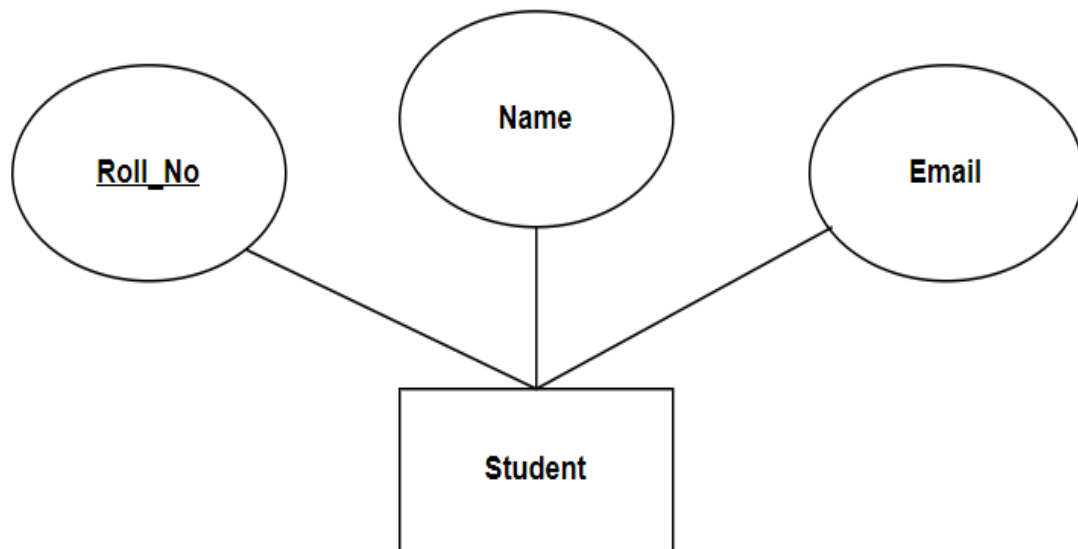
Conversion of ER diagrams to tables

Step 1 – Conversion of strong entities

- For each strong entity create a separate table with the same name.
- Includes all attributes, if there is any composite attribute divided into simple attributes and has to be included.
- Ignore multivalued attributes at this stage.
- Select the p key for the table.

Rule1: Strong Entity Set with Simple attributes

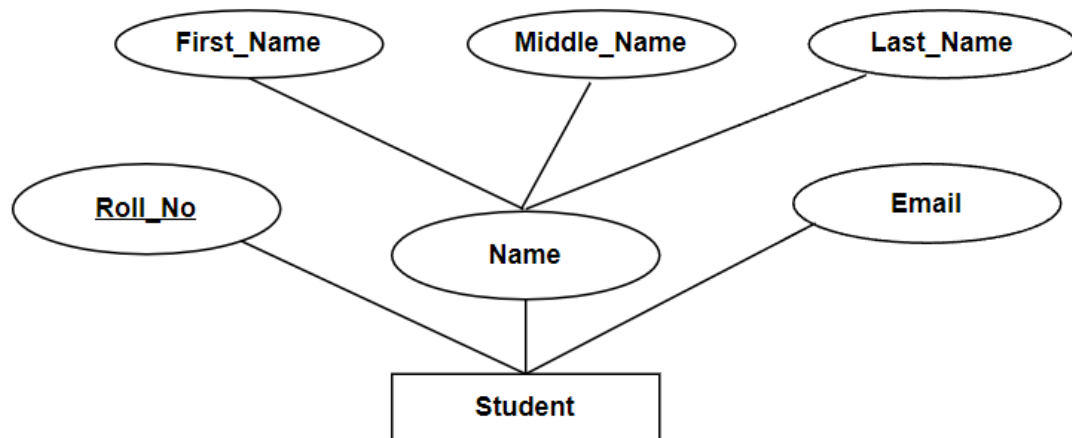
- ❑ A strong entity set with only simple attributes will require only one table in a relational model.
- ❑ **Attributes of the table** will be the **attributes of the entity set**
- ❑ The primary key of the table will be the key attribute of the entity set.



Roll_No (PK)	Name	Email

Rule2: Strong Entity Set with Composite attributes

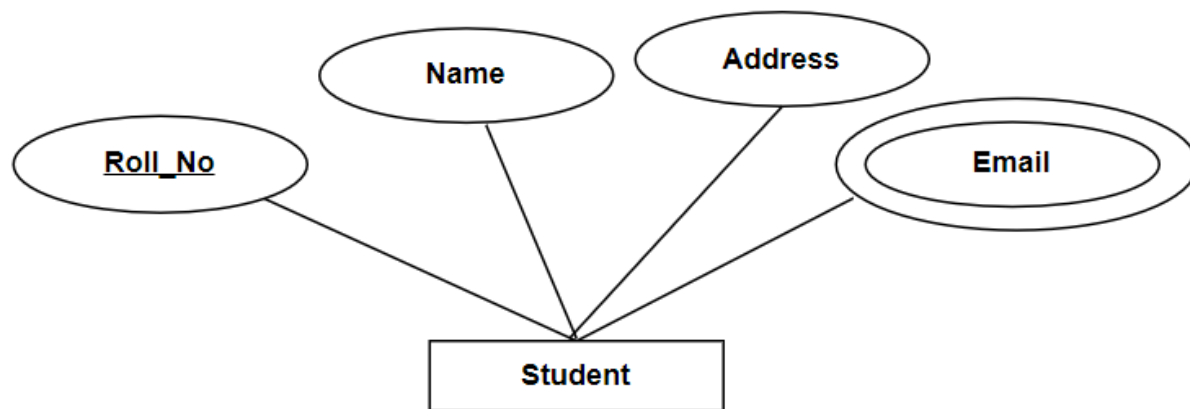
- ❑ A strong entity set with any number of composite attributes will require **only one table in relational model**.
- ❑ While conversion, simple attributes of the entity will be considered for attribute set, not the composite attribute itself.



Roll_No(PK)	First_Name	Middle_Name	Last_Name	Email

Rule3:Strong Entity Set with Multi Valued Attributes

- ❑ A **strong entity set** with any number of multi-valued attributes will **require 1+ N tables** in relational model where **N represents number of multivalued attributes**.
- ❑ One table will contain all the simple attributes with the Primary Key
- ❑ N tables will contain the Primary Key and one of the multi valued attributes (If minimum asked then 1 table only for multivalued attributes)



Roll_No (PK)	Name	Address

Roll_No (PK)	Email

Step 2 – Conversion of one-to-one relationship

- For each one to one relation, say A and B modify either A side or B side to include the P key of the other side as a foreign key.
- If A or B is having total participation, then that should be a modified table.
- If a relationship consists of attributes, include them also in the modified table.

Step 3 – Conversion of one-to-many relationship

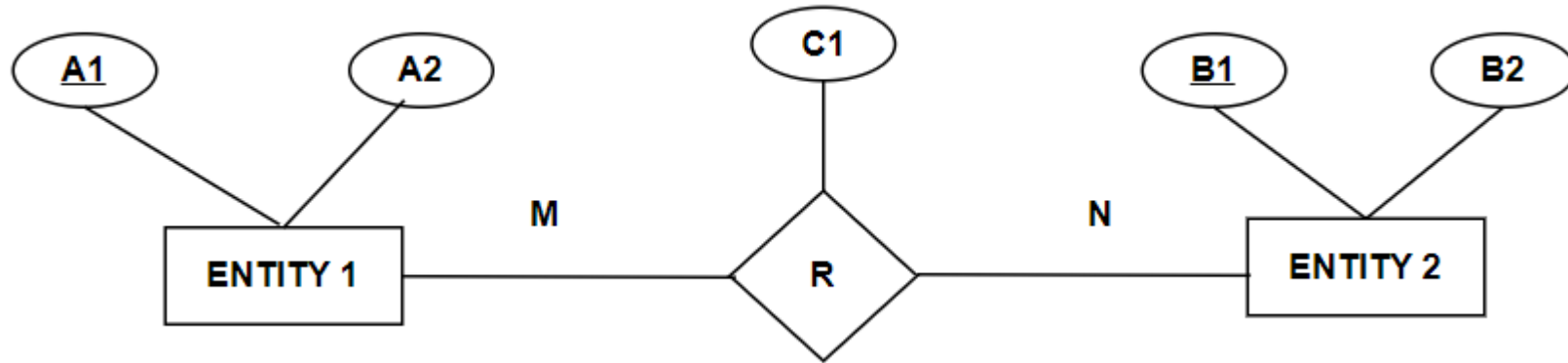
- For each one to many relationships, modify the M side to include the P key of one side as a foreign key.
- If relationships consist of attributes, include them as well.

Step 4 – Conversion of many-many relationship

- For each many-many relationship, create a separate table including the P key of M side and N side as foreign keys in the new table.
- Declare the combination of foreign keys as P for the new table.
- If relationships consist of attributes, include them also in the new table.

Rule4: Binary relationship with cardinality ratio M:N

- ❑ In many to many relationship, **three tables will be required**



A1(PK)	A2

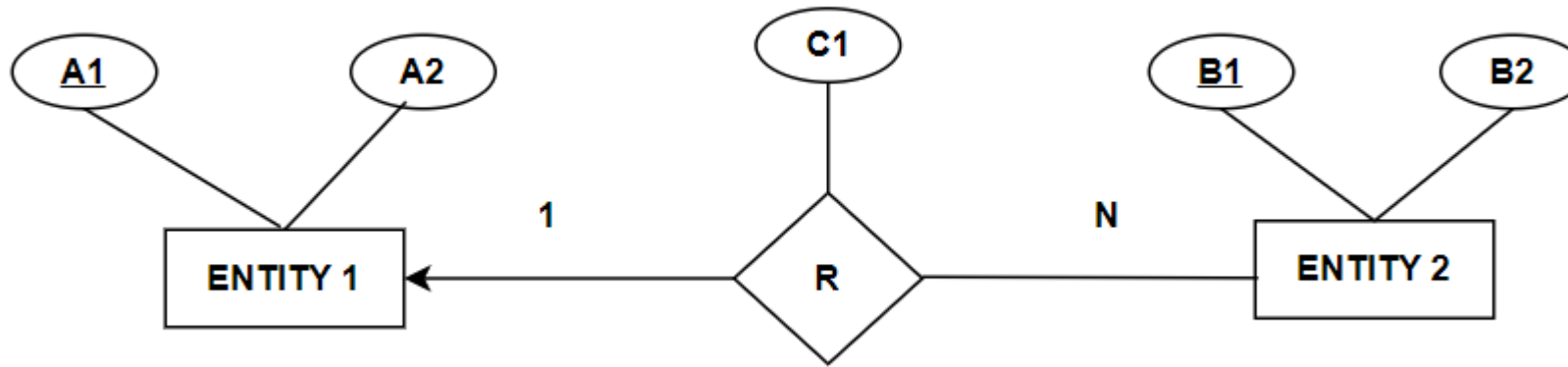
A1	B1	C1

B1(PK)	B2

PK={A1,B1}

Rule5: Binary relationship with cardinality ratio 1:N

- ❑ In **One to many** relationship **two tables** will be required

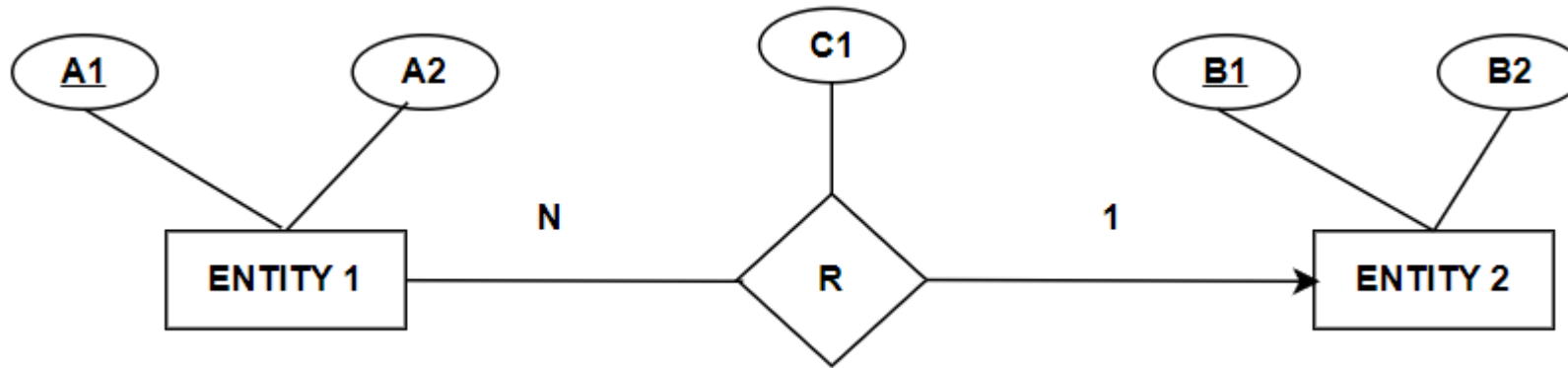


A1(PK)	A2

B1(PK)	B2	C1	A1(FK)

Rule6: Binary relationship with cardinality ratio N:1

- ❑ In many to one relationship **two tables** will be required

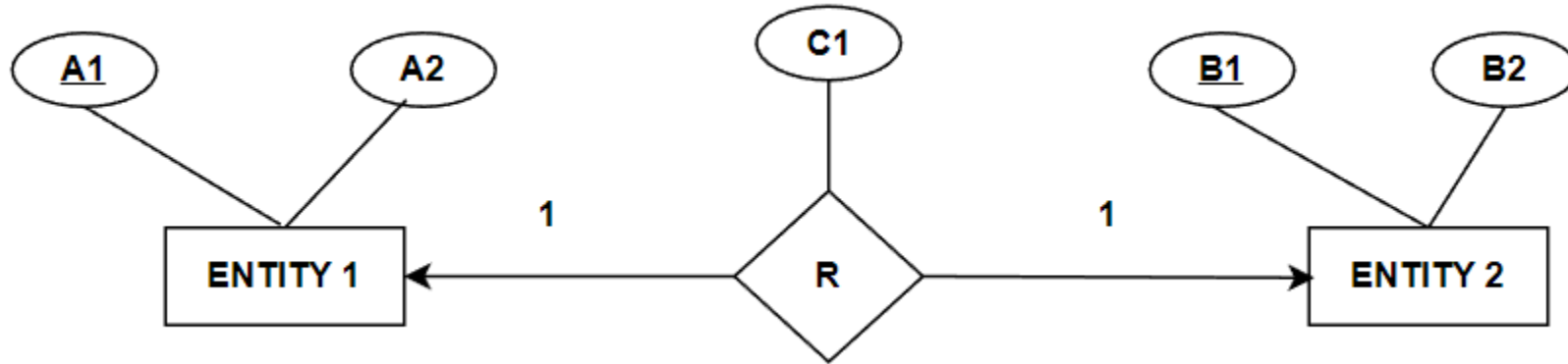


A1(PK)	A2	C1	B1(FK)

B1(PK)	B2

Rule 7: Binary relationship with cardinality ratio 1:1

- ❑ In One to One relationship two tables will be required.



A1(PK)	A2	C1	B1(FK)

B1(PK)	B2

A1(PK)	A2

B1(PK)	B2	C1	A1(FK)

Thumb Rules to Remember

While determining the minimum number of tables required for binary relationships with given cardinality ratios, following thumb rules must be kept in mind-

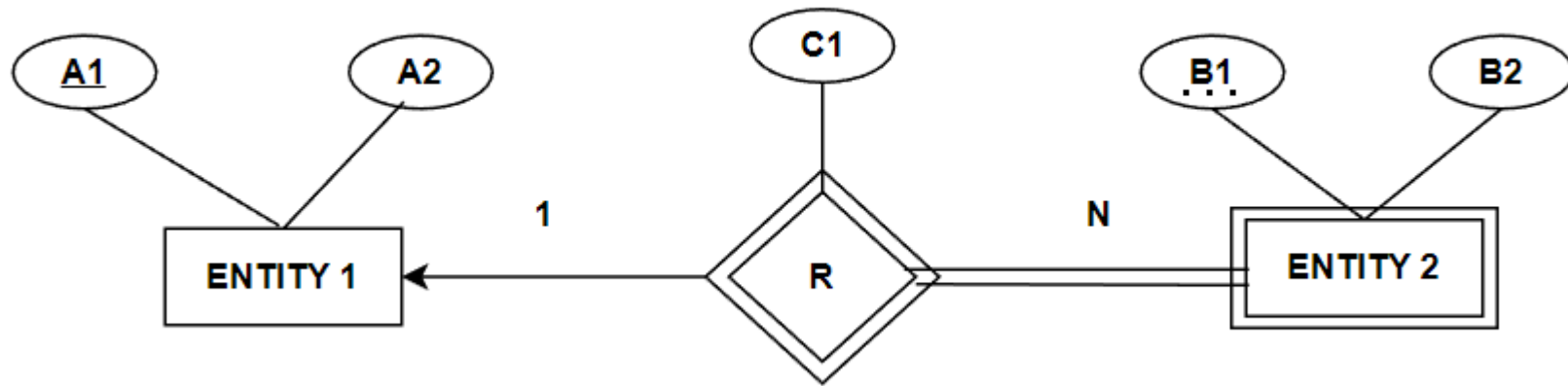
- For binary relationship with cardinality ratio $m : n$, separate and individual tables will be drawn for each entity set and relationship.
- For binary relationship with cardinality ratio either $m : 1$ or $1 : n$, always remember “many side will consume the relationship” i.e. a combined table will be drawn for many side entity set and relationship set.
- For binary relationship with cardinality ratio $1 : 1$, two tables will be required. You can combine the relationship set with any one of the entity sets.

Step 5 – Conversion of weak entity

- For each weak entity create a separate table with the same name.
- Include all attributes.
- Include the P key of a strong entity as foreign key is the weak entity.
- Declare the combination of foreign key and discriminator attribute as P key from the weak entity.

Rule 8: Binary relationship with Weak Entity Set

- ❑ The cardinality ratio between Strong and Weak Entity Set is always 1:N
- ❑ B1 is the discriminator attribute



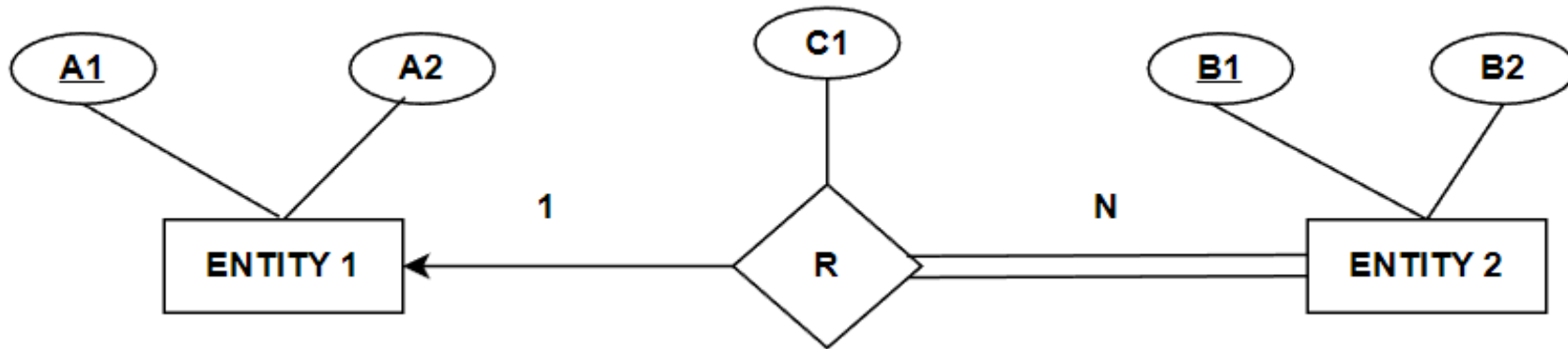
A1(PK)	A2

A1	B1	B2	C1

PK={A1,B1}

Rule 9: Binary relationship-Total participation constraint

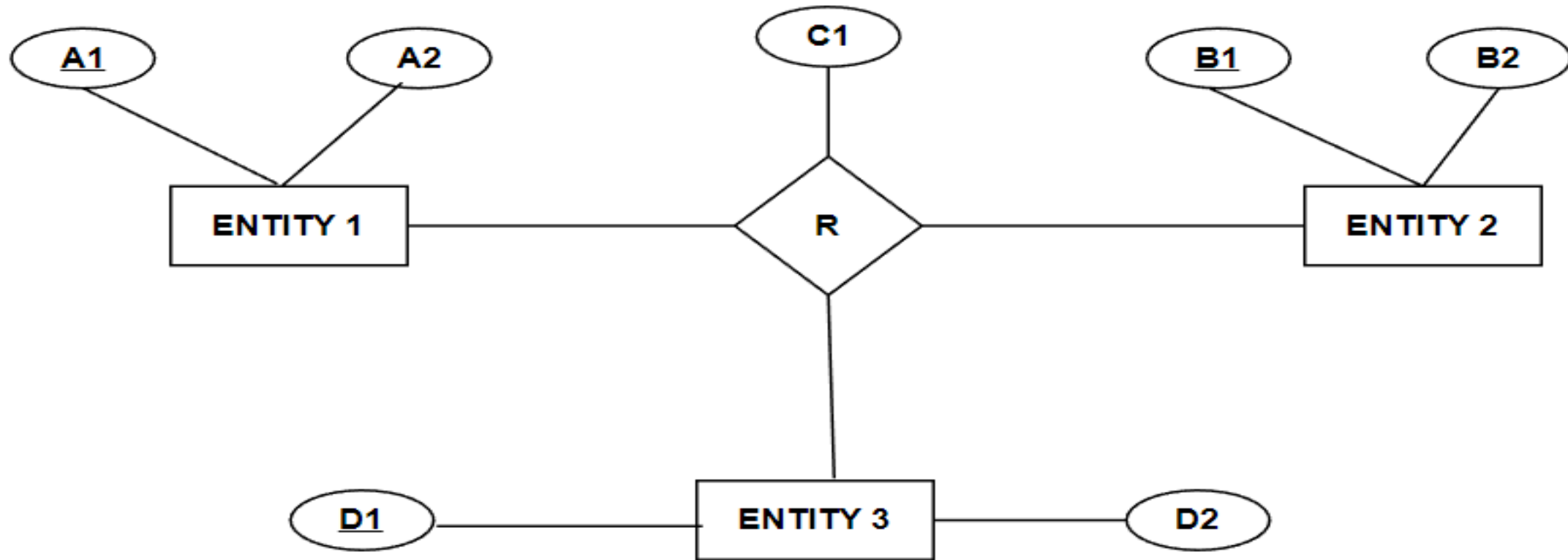
- ❑ Rule1: Strong Entity Set with Simple attributes



A1(PK)	A2

B1(PK)	B2	A1(FK-NOT NULL)	C1

Rule 10: N-Ary relationship



A1(PK)	A2

B1(PK)	B2

D1(PK)	D2

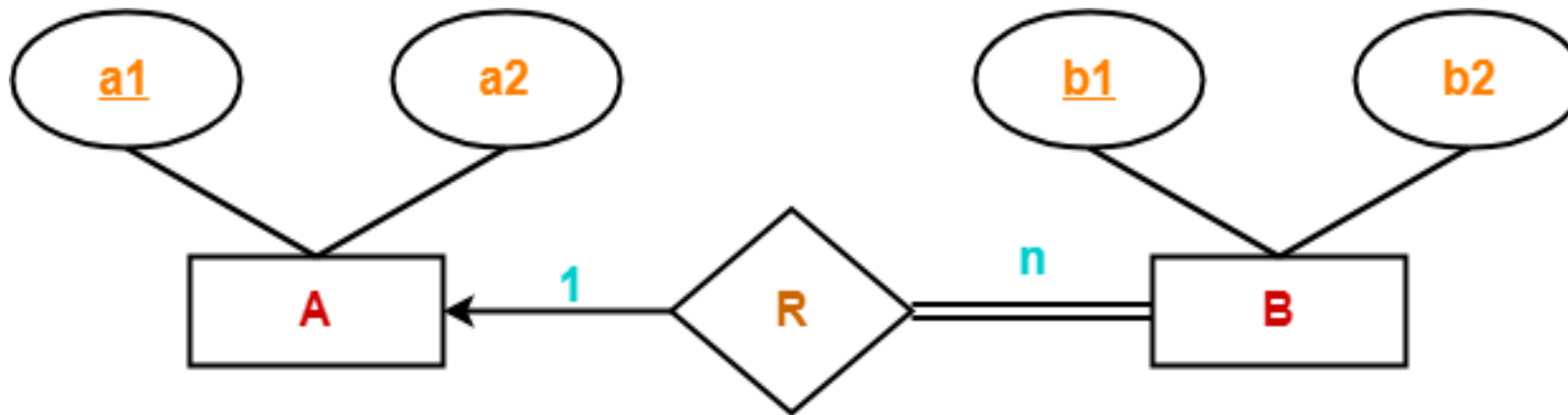
A1	B1	D1	C1

PK={A1,B1,D1}

Rule-11: For Binary Relationship With Both Cardinality Constraints and Participation Constraints-

- Cardinality constraints will be implemented as discussed in Rule-4,5,6,7.
- Because of the total participation constraint, foreign key acquires **NOT NULL** constraint i.e. now foreign key can not be null.

Case-01: For Binary Relationship With Cardinality Constraint and Total Participation Constraint From One Side



Because cardinality ratio = 1 : n , so we will combine the entity set B and relationship set R.

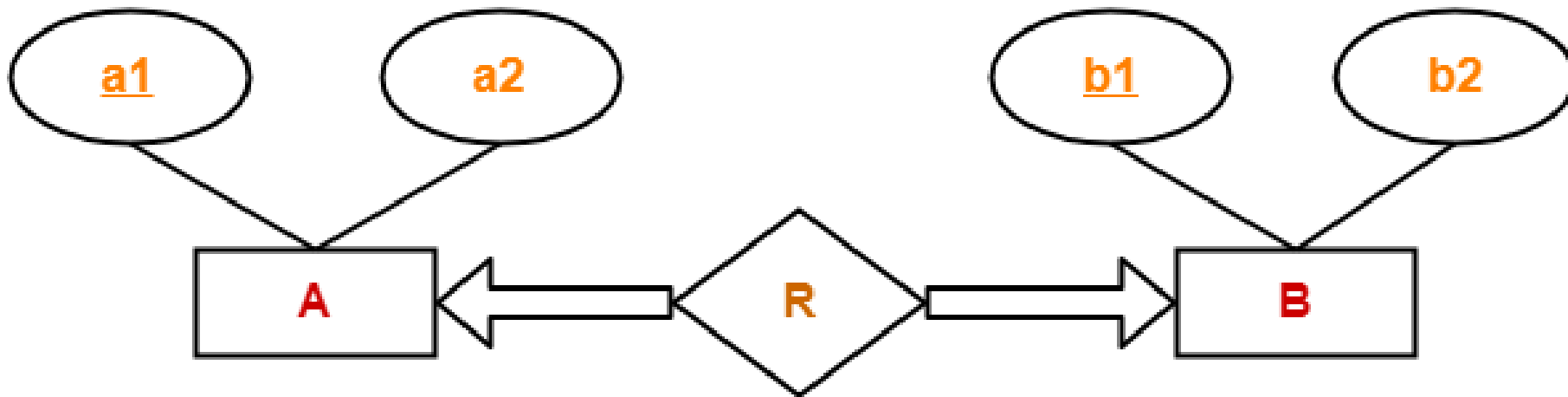
Then, two tables will be required-

1.A (a1 , a2)

2.BR (a1 , b1 , b2)

Because of total participation, foreign key a1 has acquired NOT NULL constraint, so it can't be null now.

Case-02: For Binary Relationship With Cardinality Constraint and Total Participation Constraint From Both Sides



If there is a key constraint from both the sides of an entity set with total participation, then that binary relationship is represented using only single table.

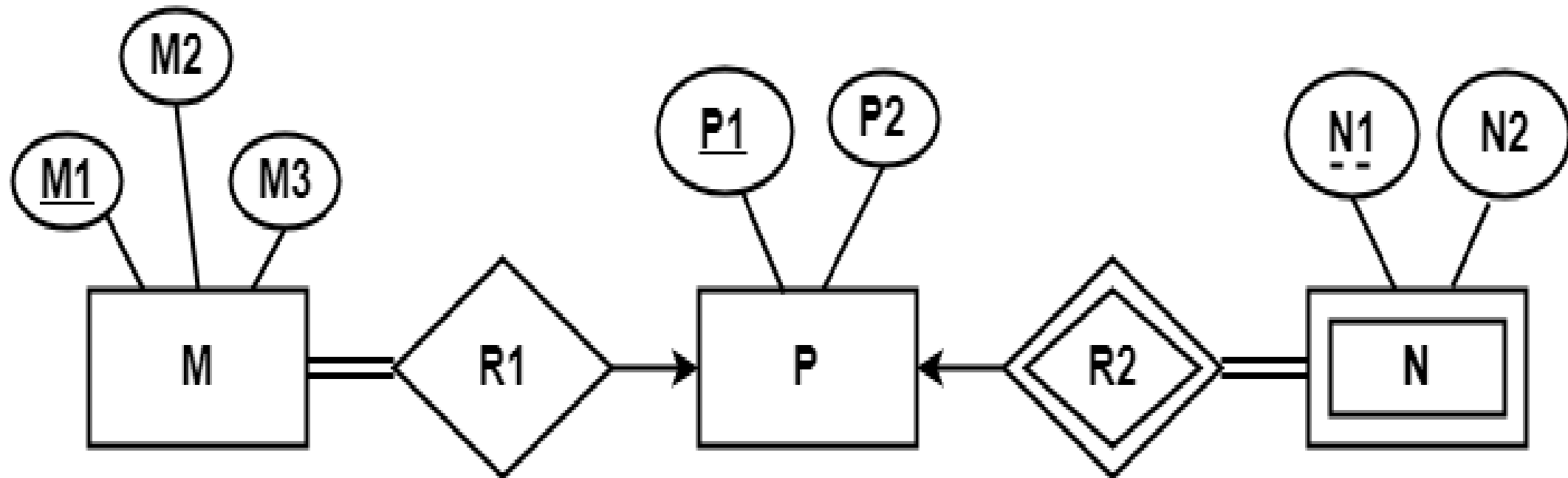
Here, Only one table is required.

ARB (a1 , a2 , b1 , b2)

**PRACTICE PROBLEMS BASED
ON CONVERTING
ER DIAGRAM TO TABLES**

Problem-01:

Find the minimum number of tables required for the following ER diagram in relational model-



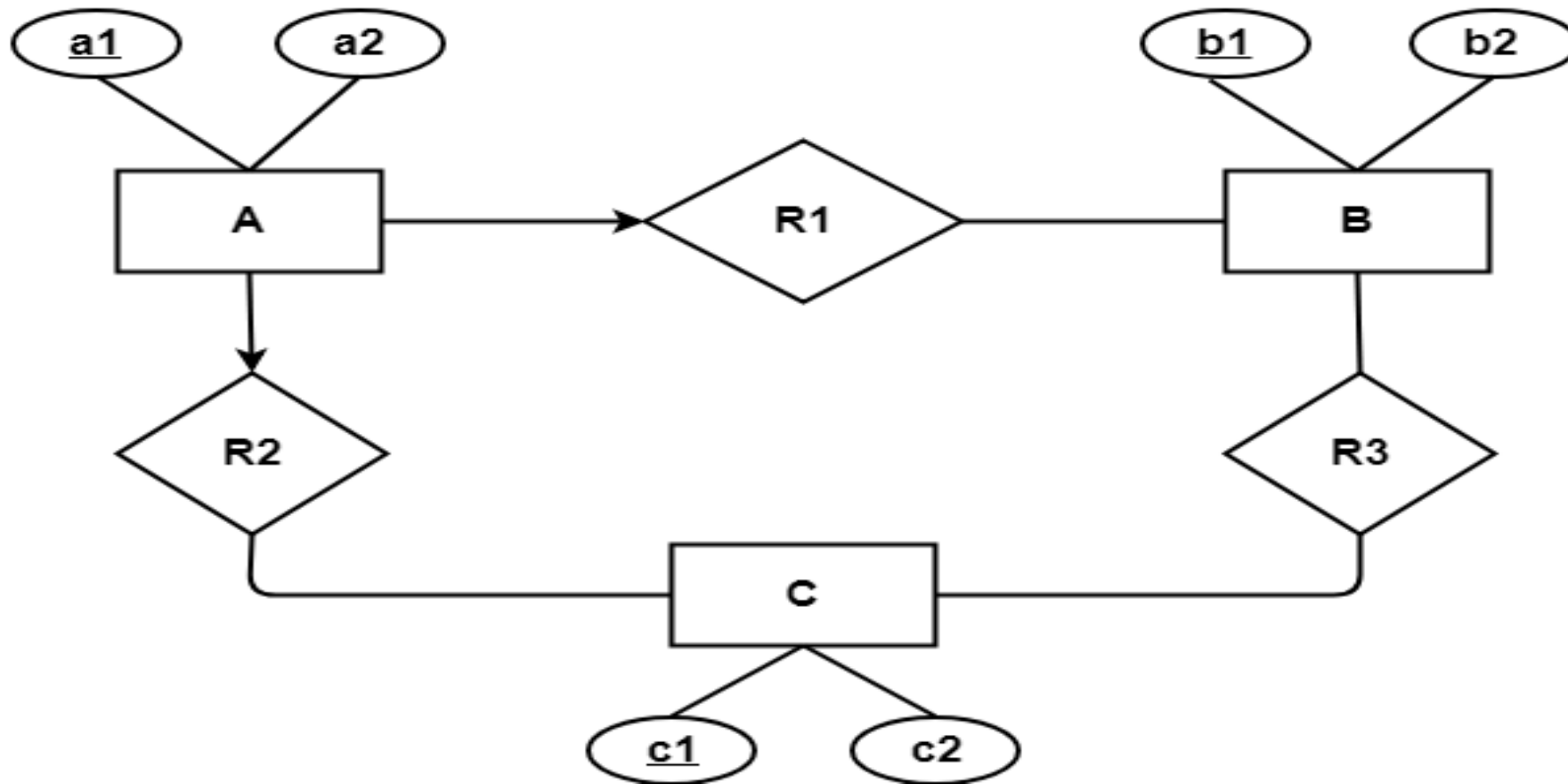
Solution-

Applying the rules, minimum 3 tables will be required-

- MR1 (M1 , M2 , M3 , P1)
- P (P1 , P2)
- NR2 (P1 , N1 , N2)

Problem-02:

Find the minimum number of tables required to represent the given ER diagram in relational model-



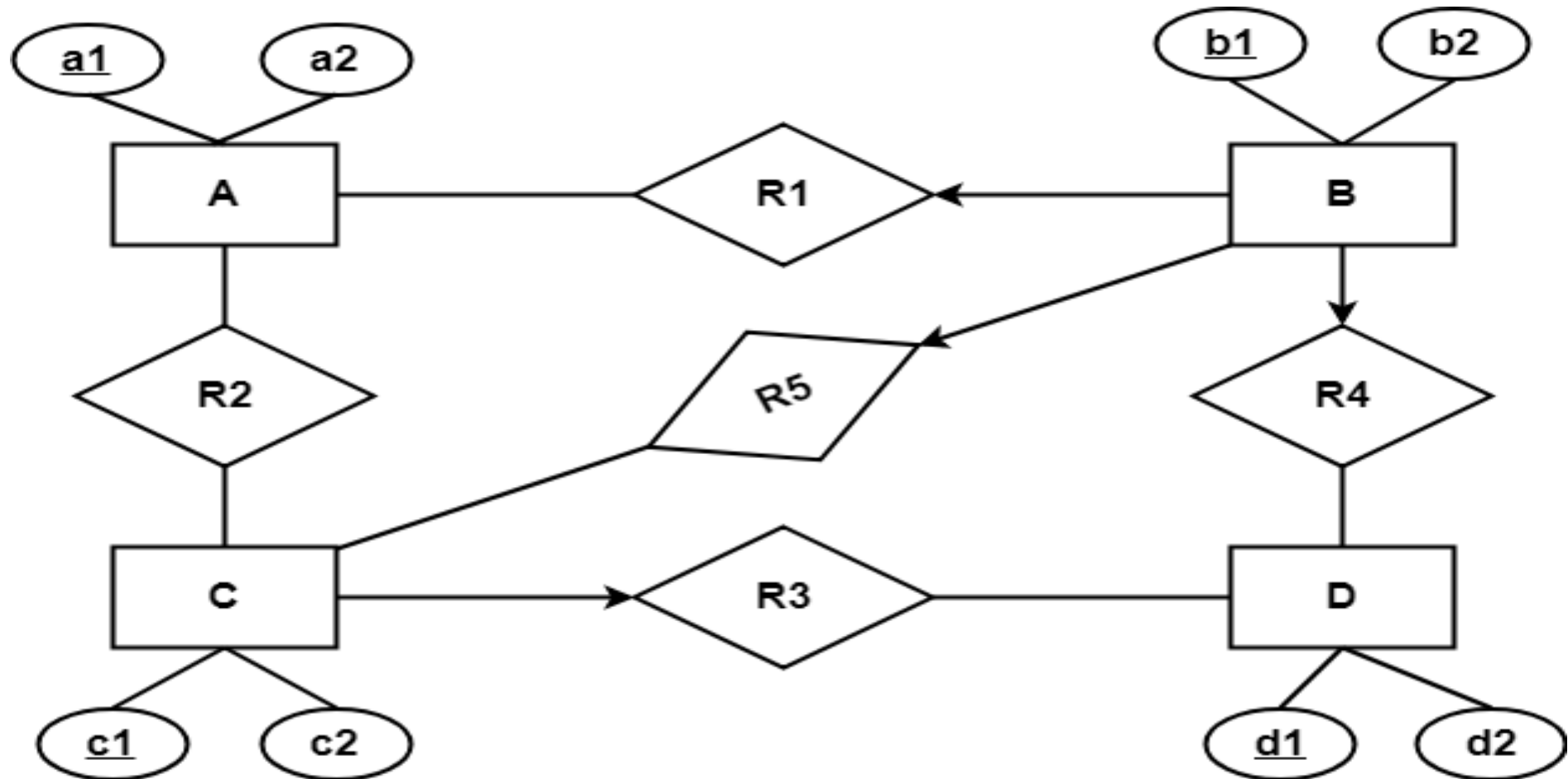
Solution-

Applying the rules, minimum 4 tables will be required-

- AR1R2 (a1 , a2 , b1 , c1)
- B (b1 , b2)
- C (c1 , c2)
- R3 (b1 , c1)

Problem-03:

Find the minimum number of tables required to represent the given ER diagram in relational model-



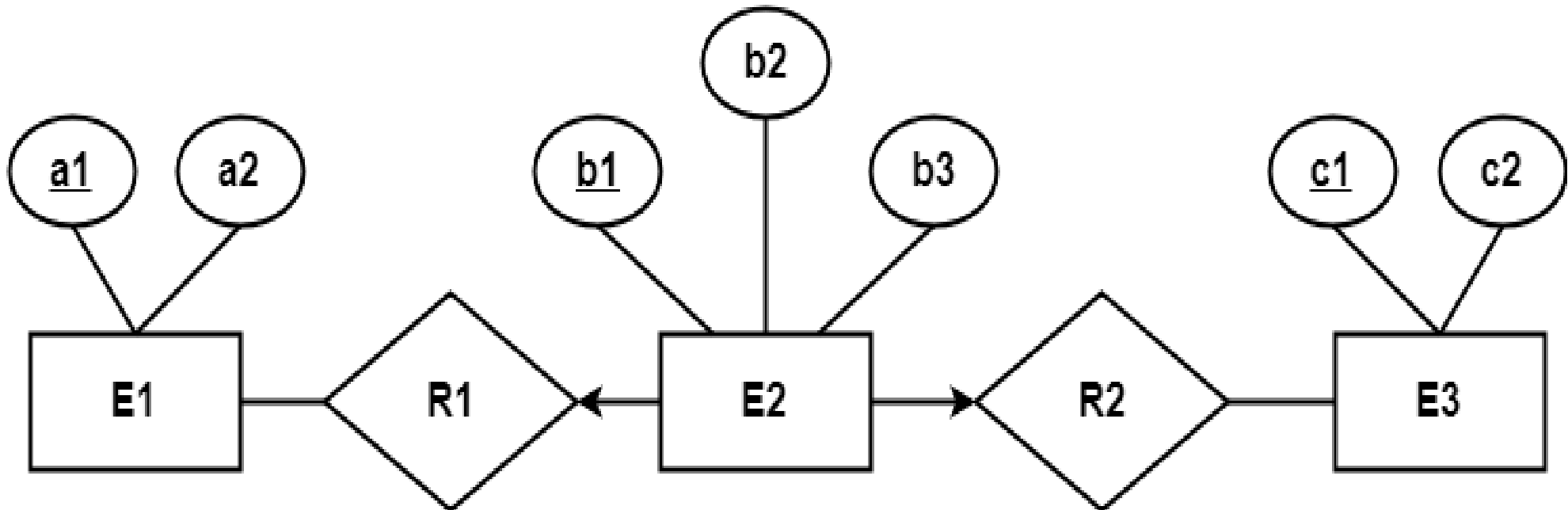
Solution-

Applying the rules, minimum 5 tables will be required-

- BR1R4R5 (b1 , b2 , a1 , c1 , d1)
- A (a1 , a2)
- R2 (a1 , c1)
- CR3 (c1 , c2 , d1)
- D (d1 , d2)

Problem-04:

Find the minimum number of tables required to represent the given ER diagram in relational model-



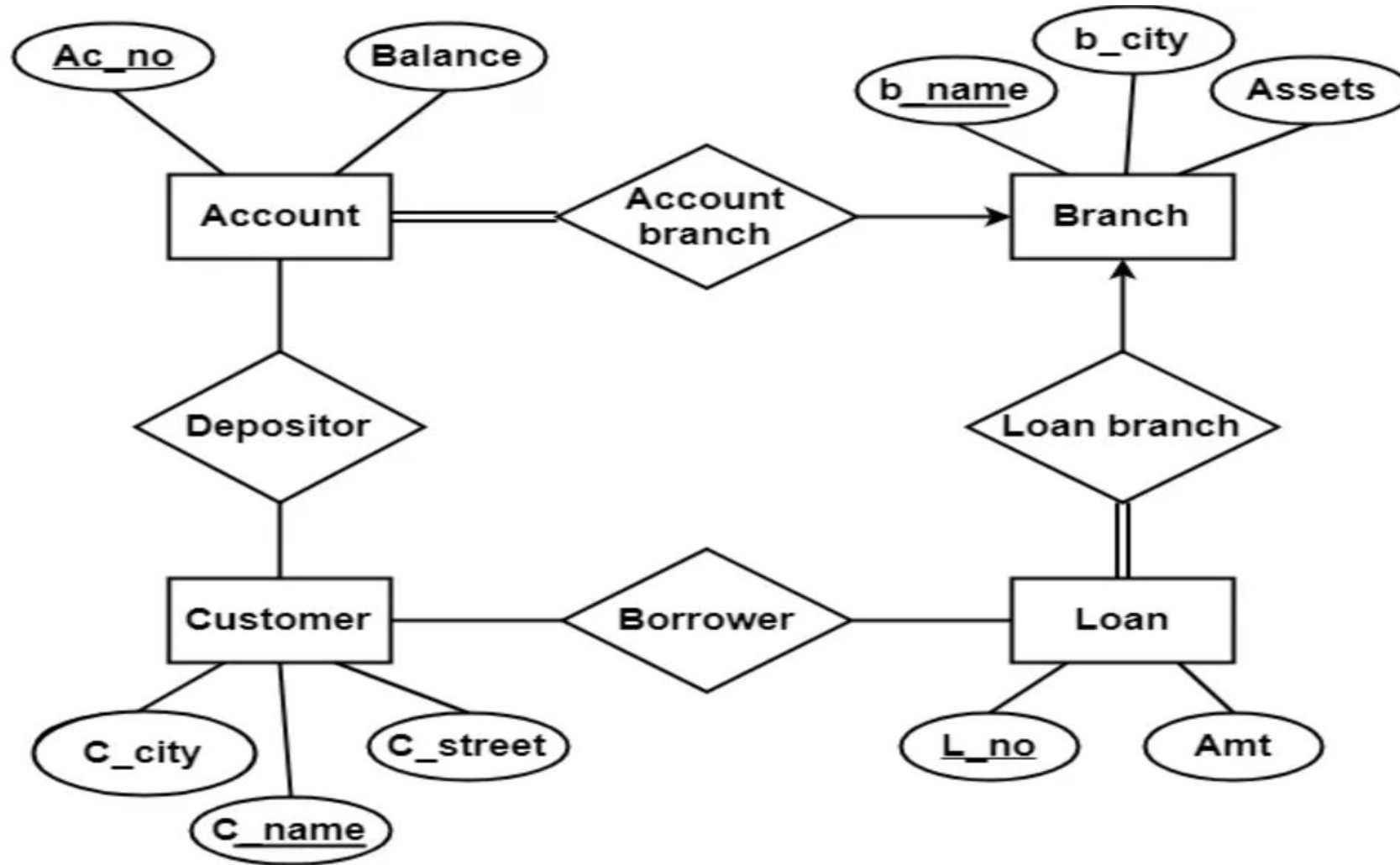
Solution-

Applying the rules, minimum 3 tables will be required-

- E1 (a1 , a2)
- E2R1R2 (b1 , b2 , a1 , c1 , b3)
- E3 (c1 , c2)

Problem-05:

Find the minimum number of tables required to represent the given ER diagram in relational model-



Solution-

Applying the rules that we have learnt, minimum 6 tables will be required-

- Account (Ac_no , Balance , b_name)
- Branch (b_name , b_city , Assets)
- Loan (L_no , Amt , b_name)
- Borrower (C_name , L_no)
- Customer (C_name , C_street , C_city)
- Depositor (C_name , Ac_no)